

UAI301 Object Oriented Programming

Lab Assignment 2

The assignment covers core topics, including basic input/output operations, string manipulation, loops, conditional statements, user-defined functions, mathematical and random libraries, exception handling, and Python's built-in data structures such as lists, dictionaries, tuples, and sets.

1: Loops

- 1) Write a Python program to print numbers from 0 to 10 using a `while` loop.
- 2) Write a Python program to print numbers from 0 to 9 using a `for` loop.
- 3) Write a Python program to print even numbers between 0 and 20 using the `range()` function.
- 4) Write a Python program to print numbers from 0 to -9 using a `for` loop.

2: Conditional Statements

- 5) Write a Python program to compare two numbers entered by the user.
- 6) Write a Python program to check whether a number is odd or even.
- 7) Write a Python program to check whether a number is prime or not.
- 8) Write a Python program to find the sum of all numbers divisible by 7 and 9 from 1 to n, where n is entered by the user.

3: Functions

- 9) Write a Python program using a function to find the sum of all odd numbers from 1 to n, where n is entered by the user.
- 10) Write a Python program using a function to find the sum of all prime numbers from 1 to n, where n is entered by the user.

4: Strings

- 11) Write a Python program to display the length, uppercase, and lowercase forms of a string.
- 12) Write a Python program to split a string using spaces and commas.
- 13) Write a Python program to check whether a string is a palindrome.

5: Random Module

- 14) Write a Python program to generate random integers between 1 and 100.
- 15) Write a Python program to generate a random password using letters and digits.
- 16) Write a Python program to generate a five-digit OTP.

6: Exception Handling

- 17) Write a Python program to handle division by zero using exception handling.
- 18) Write a Python program to handle an "array index out of range" exception.
- 19) Write a Python program to handle the "File Not Found" exception.

7: Lists

- 20) Write a Python program to create a list and display its elements.
- 21) Write a Python program to add and delete elements from a list.
- 22) Write a Python program to find the sum and average of list elements.
- 23) Write a Python program to find the maximum, minimum, and sort a list in ascending and descending order.

8: Dictionaries

- 24) Write a Python program to create a dictionary and access its values.
- 25) Write a Python program to display all keys and values of a dictionary.
- 26) Write a Python program to update, add, delete, and clear elements in a dictionary.

9: Tuples and Sets

- 27) Write a Python program to create and traverse a tuple.
- 28) Write a Python program to search for an element in a tuple.
- 29) Write a Python program to add and delete elements in a tuple by converting it to a list and back to a tuple.
- 30) Write a Python program to create a set and perform union, intersection, difference, and symmetric difference operations.
- 31) Write a Python program to add, remove, and pop elements from a set.